SOFTWARE REQUIREMENTS SPECIFICATION

Skill Development Digital Infrastructure (SDDI)

A Unified Digital Platform for Skill India, Government of Gujarat

Document Type	**Software Requirements Specification (SRS)**
Version	**1.0 (Bid Submission Draft)**
Prepared For	**Gujarat Skill Development Mission (GSDM)**
Prepared By	**AMARTRACK™ (AMAR Systems Pvt. Ltd.)**
Classification	**Confidential, Tender Response**

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Document Control

Attribute	Detail
Document Title	Software Requirements Specification, Skill Development Digital Infrastructure (SDDI)
Version	1.0, Bid Submission Draft
Status	For Tender Evaluation
Prepared By	AMARTRACK™ (AMAR Systems Pvt. Ltd.)
Intended Recipient	Gujarat Skill Development Mission (GSDM); Labour, Skill Development & Employment Department, Government of Gujarat
Confidentiality	This document contains proprietary information and is submitted solely for evaluation of the referenced procurement.

Revision History

Version	Date	Author	Summary of Change
0.1	Jul 2026	AMARTRACK	Initial draft from working prototype and stakeholder inputs
1.0	Jul 2026	AMARTRACK	Baseline for bid submission; compliance & estimation finalised

Approvals

Role	Name	Signature / Date
Prepared By (Vendor)		
Reviewed By (Technical Partner)		
Approved By (GSDM Authority)		

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1. Introduction

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1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, security, and compliance requirements for the **Skill Development Digital Infrastructure (SDDI)**, a unified digital platform proposed for the Government of Gujarat under the Skill India initiative. The platform will onboard citizens, recruit and manage trainers, deliver and track training across multiple skill sectors, manage assessment and certification, enable placement, and provide the State, districts, and administrators with real-time monitoring, reporting, and grievance redressal.

This document is the authoritative reference against which the system will be designed, developed, tested, and accepted. It is intended to be shared with the appointed technical development partner as the basis for detailed design, costing, and delivery.

1.2 Scope

The SDDI platform covers the complete skilling value chain for the State of Gujarat, across all 33 districts and multiple skill sectors including (but not limited to) Agriculture, Solar & Renewables, Welding, Network Installation, Cattle & Dairy, Computer Operator, Tele-caller/BPO, Nursing & Patient Care, Hospitality, Jewellery Design, and traditional-knowledge sectors (Astrology/Vastu, Numerology, Gemology).

In scope: citizen onboarding and identity; trainer and recruiter onboarding; role-based dashboards for State, District, Trainer, Recruiter, Citizen, and Master Admin; content library and LMS; live and recorded classes; assessment and government-verifiable certification; placement and cross-sector job discovery; grievance redressal; software helpdesk; DBT-linked stipend workflows; and comprehensive reporting, monitoring, and impact analytics.

Out of scope (current phase): hardware procurement for training centres; physical biometric device supply; the accounting/ERP system of the Department; and any module not explicitly enumerated in Section 4. Such items may be addressed via the change-control process (Section 12).

1.3 Definitions, Acronyms & Abbreviations

Term	Definition
SDDI	Skill Development Digital Infrastructure, the platform specified herein
GSDM	Gujarat Skill Development Mission
DPI	Digital Public Infrastructure
India Stack	Set of open APIs, Aadhaar, DigiLocker, e-Sign, UPI, Account Aggregator
DBT	Direct Benefit Transfer (via PFMS/NPCI APB)
e-KYC	Electronic Know Your Customer (Aadhaar-based)
RBAC	Role-Based Access Control
NCVET	National Council for Vocational Education and Training
APAAR	Automated Permanent Academic Account Registry
ABC	Academic Bank of Credits
DPDP	Digital Personal Data Protection Act, 2023
CERT-In	Indian Computer Emergency Response Team
VAPT	Vulnerability Assessment & Penetration Testing
MeitY	Ministry of Electronics & Information Technology
SLA	Service Level Agreement
LMS	Learning Management System
SSO	Single Sign-On
KPI	Key Performance Indicator

1.4 Intended Audience

This SRS is intended for: the GSDM tender evaluation committee and technical reviewers; the empanelling authority within the Labour, Skill Development & Employment Department; the appointed system integrator / development partner; the vendor's project, QA, and security teams; and third-party auditors (CERT-In empanelled) conducting VAPT and compliance review.

1.5 References & Compliance Basis

- Digital Personal Data Protection Act, 2023 (DPDP Act) and rules thereunder.
- Aadhaar Act, 2016 and UIDAI e-KYC / authentication guidelines.
- MeitY Guidelines for empanelment of Cloud Service Providers (GI Cloud / MeghRaj) and data localization norms.
- CERT-In guidelines on security auditing, incident reporting, and log retention.
- NCVET / NSQF frameworks for skilling qualifications and certification.
- DigiLocker, APAAR, ABC, and DBT/PFMS integration specifications (via API Setu).
- WCAG 2.1 AA accessibility guidelines and Government of India accessibility standards (GIGW).

2. Overall Description

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2.1 Product Perspective

SDDI is a new, self-contained Digital Public Infrastructure product for the State of Gujarat. It is designed as an interoperable platform that consumes national DPI services (Aadhaar e-KYC, DigiLocker, DBT, APAAR/ABC) and exposes State-level skilling services to citizens, trainers, recruiters, and government officers. It replaces fragmented, manual, and paper-based skilling workflows with a single monitored digital backbone.

2.2 Product Functions

At the highest level, the platform shall provide: (a) citizen identity, onboarding and lifecycle tracking; (b) trainer recruitment, batch and attendance management; (c) content delivery and live/recorded classes; (d) assessment and government-verifiable certification; (e) placement and cross-sector opportunity discovery; (f) State/District dashboards, reporting, and impact monitoring; (g) grievance redressal with SLA tracking; (h) a software helpdesk and platform administration console; (i) DBT-linked stipend and benefit workflows; and (j) mobile access with offline-first field capability.

2.3 User Classes & Characteristics (RBAC)

User Class	Primary Responsibilities	Characteristics
State Skill Mission / SOP	Statewide policy, oversight, SOP configuration, escalations, statewide analytics.	High digital literacy; few users; read-most, configure-some.
District Skill Officer	District operations, citizen registry, trainer management, document validation, local grievances.	Moderate literacy; ~33+ district offices; high transactional use.
Trainer	Batch delivery, attendance, live classes, assessments, content access.	Variable literacy; thousands of users; mobile-heavy, often rural.
Recruiter / Hiring Partner	Certified talent pool access, job postings, shortlisting, hiring insights.	High literacy; hundreds of empanelled employers.
Citizen / Trainee	Registration, training, documents, certificates, jobs, grievances, advancement.	Widely variable literacy; millions of users; mobile-first, low-bandwidth, multilingual.
Software Master Admin	Platform administration, helpdesk, user/role management, audit, system config.	Vendor/ops team; very few; full privileged access with audit.
Tele-Customer Service Agent	Inbound/outbound helpline; citizen query resolution, follow-up and verification calls; grievance/ticket logging on behalf of citizens; call-disposition and CRM updates.	Contact-centre staff; moderate literacy; script-driven; multilingual (Gujarati/Hindi/English).
MIS Team (Hub & Spoke)	Central MIS hub consolidating data from district/zone spokes; SOP and target dissemination downward; report integrity, data reconciliation, and consolidated statewide MIS.	Central hub + zonal spokes; high literacy; reporting-heavy; controlled data-edit rights.
Onsite Supervisor (Field)	Centre-level verification that classes run as scheduled; learner presence and infrastructure checks; geo-tagged field reports feeding the monitoring layer; raised as-needed per centre/batch.	Field staff; mobile-first, offline-capable; deployed as-needed by district/zone.
Nodal QC & SOP-Adherence Team	Quality-control and compliance cell; audits SOP conformance across the hub-and-spoke network; QC sampling of batches/trainers/centres; raises and tracks non-conformance to closure.	Central + zonal nodal officers; high literacy; audit & escalation authority.
Content & Creative Team	Produces and manages training content, creative visuals, infographics, multilingual literature, animations, and video courses; content versioning, localisation (Gujarati/Hindi/English), review & publishing workflow into the Content Library/LMS.	Central studio team; high literacy; content-authoring & media-upload rights with editorial approval workflow.
Data Science Team	Owns analytics, impact measurement, and predictive models, dropout-risk, placement-propensity, demand-forecasting, and anomaly/fraud detection; maintains dashboards, KPIs, and data pipelines feeding State/District monitoring.	Small central team; high literacy; read access to anonymised/aggregated data under DPDP governance; model-deployment rights.

2.4 Operating Environment

The platform shall operate as a cloud-hosted, browser-accessible web application with companion Android/iOS mobile applications. It shall be hosted exclusively on **MeitY-empanelled cloud infrastructure within India** (GI Cloud / MeghRaj or empanelled CSP, India region only). Client environments include modern browsers (Chrome, Edge, Firefox, Safari), Android 9+ and iOS 14+ devices, and shall degrade gracefully on low-bandwidth 2G/3G rural connectivity.

2.5 Design & Implementation Constraints

- **Data localization:** all citizen and government data shall reside within India; no data egress to foreign jurisdictions.
- **Compliance-first:** DPDP Act, Aadhaar Act, and CERT-In requirements are mandatory constraints, not options.
- **Interoperability:** integrations shall use published India Stack / API Setu specifications and open standards (REST/JSON, OAuth2, FHIR where applicable).
- **Accessibility & language:** UI shall support Gujarati, Hindi, and English, and meet WCAG 2.1 AA / GIGW.
- **Offline-first field layer:** attendance and content shall function offline and sync on reconnection.

2.6 Assumptions & Dependencies

- UIDAI, DigiLocker, DBT/PFMS, and APAAR/ABC sandbox and production access will be granted to the empanelled vendor by the respective authorities.
- The Department will nominate a data fiduciary / consent manager and approve SOP thresholds.
- Third-party job-board and course-platform feeds (where used) are subject to the respective providers' commercial API terms.
- Live-class video is delivered via an integrated conferencing service (self-hosted open-source or licensed SDK), not built in-house.

3. System Architecture

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3.1 Architecture Overview

SDDI shall be implemented as a **modular monolith with a service-oriented internal structure**, deployed on MeitY-empanelled Indian cloud infrastructure. A modular monolith is recommended over premature microservices to balance delivery speed, operational simplicity, and cost, while preserving clean module boundaries that can be extracted into independent services as load grows. The architecture is organised into four logical tiers: (1) a presentation tier (responsive web + Android/iOS apps with an offline-first field layer); (2) an API/application tier exposing secured REST/JSON services with OAuth2/JWT and RBAC enforcement; (3) a data tier (relational primary store, cache, object storage, and an analytics/reporting store); and (4) an integration tier connecting to India Stack and third-party services via API Setu.

3.2 Recommended Technology Stack

The following stack is recommended on the basis of maintainability, Indian hiring availability, security posture, and suitability for government workloads. Final selection rests with the appointed development partner, subject to the constraints in Section 2.5.

Layer	Recommended Choice	Rationale
Web Frontend	React with Next.js (SSR)	Large Indian talent pool; server-side rendering aids low-bandwidth rural access; component reuse from prototype.
Mobile	React Native or Flutter	Single codebase for Android/iOS; offline-first field capability for attendance & content.
Backend / API	Django (Python) or NestJS (Node.js)	Batteries-included RBAC, admin, and auth (Django); or single-language JS stack (NestJS).
Primary Database	PostgreSQL	Relational integrity for citizen→trainer→certificate→placement graph; auditable; strong support.
Cache / Sessions	Redis	Session store, rate-limiting, and hot-path caching.
Object Storage	S3-compatible (India region)	Document vault, certificates, media; encrypted at rest.
Analytics Store	Columnar warehouse (e.g. PostgreSQL + read replicas / OLAP)	Separates reporting load from transactional DB; feeds Data Science pipelines.
Live Classes	Integrated conferencing (self-hosted Jitsi or licensed SDK)	Data-localisable video; not built in-house.
Search	OpenSearch / Elasticsearch	Global search across citizens, trainers, content, tickets.
Infrastructure	MeitY-empanelled cloud (GI Cloud / CSP, India only)	Data localization; CERT-In-auditable; horizontally scalable.
CI/CD & Ops	Containerised (Docker) + orchestration; IaC	Repeatable deploys; audit-friendly; blue-green releases.

> **Architecture principle.** Start as a modular monolith; extract services (e.g. notifications, live-class, analytics) only where load or team boundaries justify it. This is a deliberate risk-reduction choice for a population-scale first release.

3.3 India Stack & DPI Integration

SDDI shall integrate national Digital Public Infrastructure through published specifications (via API Setu) rather than bespoke connectors:

DPI Service	Integration Purpose
Aadhaar e-KYC / Authentication	Citizen & trainer identity verification at onboarding; consent-based, UIDAI-compliant; no storage of raw Aadhaar beyond permitted reference.
DigiLocker	Issue and store government-verifiable certificates; fetch verified documents (Aadhaar, PAN, education) with citizen consent.
DBT / PFMS (APB)	Aadhaar-seeded stipend and scholarship disbursement; transaction status reconciliation.
APAAR / ABC	Academic account and credit linkage for skilling credits and NEP alignment.
e-Sign	Digital signing of certificates, consent forms, and official approvals.
Consent Framework (DPDP)	Consent capture, purpose limitation, and revocation for all personal-data processing.

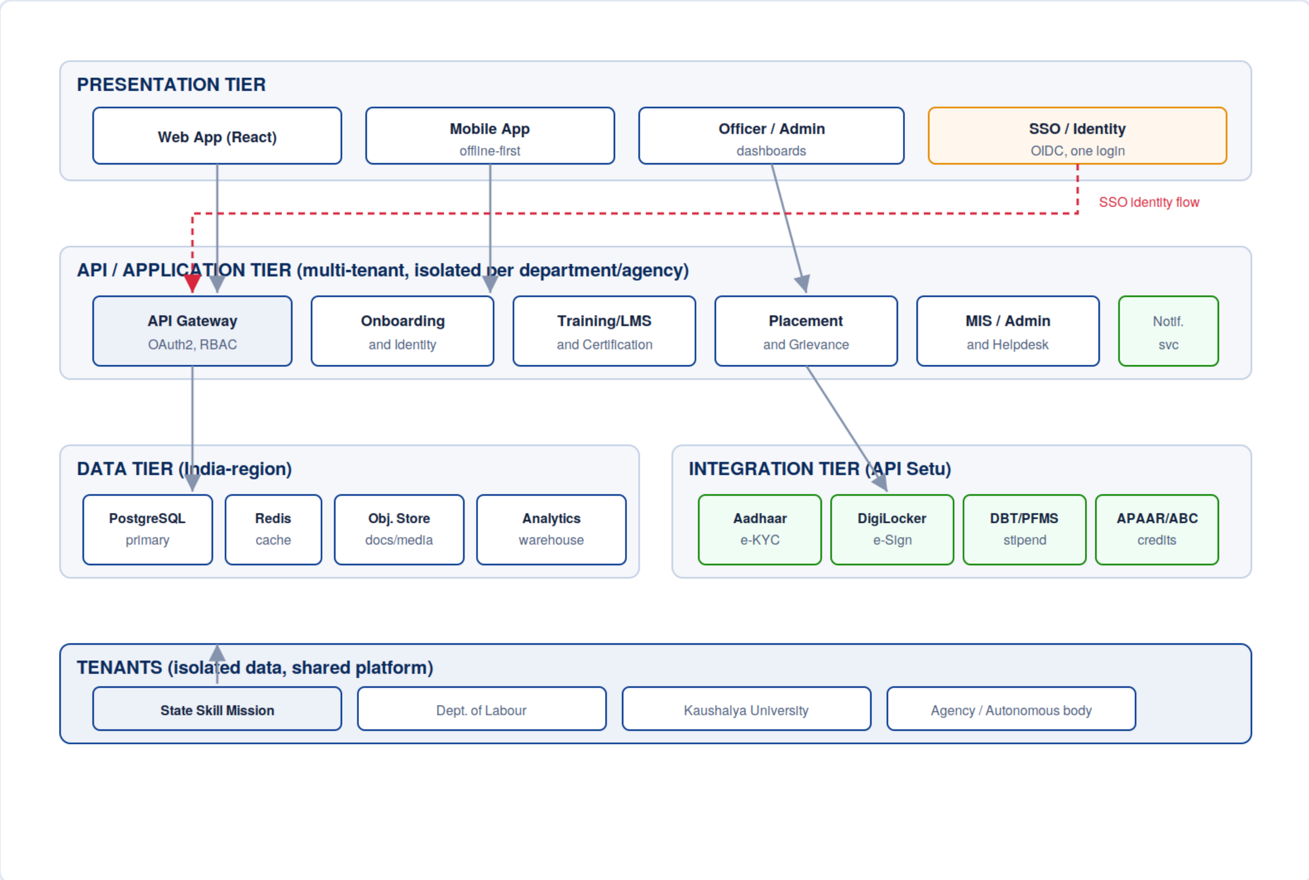
3.4 Data Architecture & Hub-and-Spoke MIS Model

The platform's data and reporting model follows a **hub-and-spoke architecture** that mirrors the operational structure of the skilling programme. The **central MIS hub** holds the consolidated statewide dataset and disseminates SOPs, targets, and configuration downward. **Spokes** (zonal, North/South/East/West, and district nodes) capture ground-level transactions (onboarding, attendance, assessments, field reports) that are consolidated upward in near-real-time.

Data flows are bidirectional and governed: transactional data flows up from centres→districts→zones→State hub; policy, SOP thresholds, and targets flow down the same spine. The **Nodal QC team** and **Onsite Supervisors** inject verification/ground-truth data at the spoke level, while the **Data Science team** consumes anonymised aggregates at the hub for impact analytics. This structure ensures a single source of truth at the hub while preserving spoke-level operational autonomy and auditability.

3.5 System Architecture Diagram

The following logical view shows the four core tiers, presentation, API/application, data, and integration, together with the India Stack integration layer and the platform’s multi-tenant structure. The platform is deployed once and serves multiple tenants (the State Skill Mission and its participating departments, agencies, and autonomous bodies), each with isolated data, configuration, and branding on a shared, centrally-maintained deployment.



Solid arrows: standard data/service flows. Dashed red: single sign-on (SSO) identity flow, one authenticated identity resolves across all modules and authorised tenants via the central identity provider.

3.6 Single Sign-On (SSO) & Identity

SDDI shall implement **standards-based single sign-on** so that a user authenticates once and accesses every module and authorised tenant for which they hold a role, without re-logging in as they move between onboarding, training, assessment, placement, grievance, MIS, and helpdesk. A single citizen, trainer, officer, or administrator therefore has one identity and one login spanning the platform's twelve user roles, which simplifies the user experience, reduces password fatigue and support load, and gives a single, auditable point of authentication and access control.

3.6.1 Design

- **Protocol:** OpenID Connect (OIDC) over OAuth2 for citizens/officers; SAML 2.0 supported for government directory integration where required.
- **Identity anchor:** Aadhaar e-KYC (consent-based) establishes a verified identity; the platform stores only permitted reference tokens, never raw Aadhaar.
- **Token model:** short-lived JWT access tokens with rotating refresh tokens; scopes encode role (RBAC) and tenant claims.
- **Central identity provider:** a single identity provider issues and validates tenant-scoped assertions, so a department or agency tenant can trust an identity verified centrally without duplicating identity data.
- **MFA:** mandatory multi-factor authentication for all privileged roles (State, District, Master Admin, Nodal QC, Cybersecurity).
- **Session & audit:** centralised session management with single logout; every authentication and cross-tenant assertion is logged immutably (user, tenant, IP, timestamp, geo).

> **Why SSO matters.** A single sign-on across the platform's twelve roles and every module means a citizen, trainer, officer, or administrator authenticates once and moves seamlessly between onboarding, training, assessment, placement, grievance, MIS, and helpdesk. This reduces friction and support load, removes duplicate identities, and gives the State one auditable, centrally-governed point of authentication and access control.

3.7 Multi-Tenancy & Data Isolation

The platform is architected from day one as **multi-tenant**: a single, centrally-maintained deployment serves the State Skill Mission and its participating departments, agencies, and autonomous bodies (for example the Department of Labour, Skill Development & Employment and Kaushalya Skill University), each as an isolated tenant with its own data, configuration, branding, and language. This is a deliberate design choice so that bringing a new department or agency onto the platform is a configuration and onboarding exercise rather than a fresh build, while every tenant's data stays cleanly separated.

3.7.1 Tenancy & Isolation Model

- **Isolation model:** logical isolation with per-tenant data segregation (row/schema-level) and, where a tenant requires it, physical database separation, so each tenant's data remains its own.
- **Localisation:** per-tenant sectors, SOP thresholds, schemes, language, and branding without code changes.
- **Data fiduciary:** the State (and the relevant tenant body) is the data fiduciary for its citizens under the DPDP Act; the platform operator acts as a processor under agreement.

3.7.2 Data Localisation & Sovereignty

All tenant data resides within India on MeitY-empanelled infrastructure. Tenant boundaries are enforced at the application, data, and audit layers so that access, reporting, and analytics are always scoped to the tenant's own data unless an explicit, consented, and logged cross-tenant sharing agreement is in place within the State. This preserves a single, coherent platform while keeping each department's or agency's data isolated, auditable, and under its own governance.

4. Functional Requirements

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Requirements are identified as `FR-x.y` and prioritised using MoSCoW: **MUST**, **SHOULD**, **COULD**. Each requirement is testable and traceable to acceptance criteria (Section 14).

4.1 Identity, Onboarding & Profile

Req ID	Requirement	Priority	Role(s)
`FR-1.1`	The system shall allow citizens to self-register via a multi-step flow capturing personal details, gender, age, mobile, and photograph.	**MUST**	Citizen
`FR-1.2`	The system shall verify identity via Aadhaar e-KYC with explicit, revocable consent.	**MUST**	Citizen, Trainer
`FR-1.3`	The system shall maintain a digital identity locker (document vault) for Aadhaar, PAN, EPIC, APAAR, ABC, PEN, Ration Card, and DBT bank account, each with a verification state.	**MUST**	Citizen, District
`FR-1.4`	The system shall capture and store consent for programme participation and data-sharing (programme-vs-geography consent).	**MUST**	Citizen
`FR-1.5`	The system shall generate a unique ID (UID) and a printable QR-enabled ID/admit card per user.	**MUST**	Citizen, Trainer
`FR-1.6`	The system shall allow linking of family-member IDs for household-scheme mapping.	**SHOULD**	Citizen
`FR-1.7`	The system shall display member-since, verification status, and acquired certificates on the profile.	**SHOULD**	Citizen

4.2 Training Lifecycle Management

Req ID	Requirement	Priority	Role(s)
`FR-2.1`	The system shall track each citizen through a defined lifecycle: Registered → In Training → Assessment → Certified → Placed.	**MUST**	Citizen, District, State
`FR-2.2`	The system shall support enrolment into sector programmes and batches, with eligibility checks (course/job/scheme/advancement).	**MUST**	Citizen, District
`FR-2.3`	The system shall allow officers to advance, hold, or flag a citizen's lifecycle stage with reason codes.	**MUST**	District, State
`FR-2.4`	The system shall provide a peer-comparison view filtered by age/gender/sector for motivational benchmarking.	**COULD**	Citizen

4.3 Trainer & Batch Management

Req ID	Requirement	Priority	Role(s)
FR-3.1	The system shall onboard and empanel trainers with certification (NCVET) and specialisation records.	MUST	District, State
FR-3.2	The system shall manage batches, rosters, schedules, and trainer assignment.	MUST	Trainer, District
FR-3.3	The system shall record trainer hours delivered, attendance participation, and ratings.	MUST	Trainer, District, MIS
FR-3.4	The system shall flag trainer attendance anomalies (e.g. marked present without session logs).	SHOULD	Nodal QC, District

4.4 Content Library & LMS

Req ID	Requirement	Priority	Role(s)
FR-4.1	The system shall provide a sector-filterable content library of videos, PDFs, and infographics.	MUST	All
FR-4.2	The Content & Creative team shall author, version, localise (Gujarati/ Hindi/English), review, and publish content through an editorial approval workflow.	MUST	Content Team
FR-4.3	The system shall support creative visuals, animations, and multilingual literature as first-class content types.	SHOULD	Content Team
FR-4.4	The system shall track content consumption (views, completion) per learner and per module.	SHOULD	MIS, Data Science
FR-4.5	The system shall serve cached content offline on the mobile field layer.	SHOULD	Citizen, Trainer

4.5 Live Classes & Attendance

Req ID	Requirement	Priority	Role(s)
FR-5.1	The system shall host live online classes via integrated conferencing (Google Meet/Zoom/Jitsi) with admin-shared join links.	MUST	Trainer, Citizen
FR-5.2	Qualified administrators (State/District/Nodal QC) shall be able to drop in to monitor any live class.	SHOULD	State, District, Nodal QC
FR-5.3	The system shall record classes and archive them with IP address, server timestamp, and geo-tag for audit.	MUST	Master Admin, Nodal QC
FR-5.4	The system shall support attendance marking (in-person and online), functioning offline and syncing on reconnection.	MUST	Trainer, Onsite Supervisor
FR-5.5	Onsite Supervisors shall submit geo-tagged field verification that a class occurred as scheduled.	SHOULD	Onsite Supervisor

4.6 Assessment & Certification

Req ID	Requirement	Priority	Role(s)
FR-6.1	The system shall manage assessments and record results against modules and batches.	MUST	Trainer, District
FR-6.2	The system shall issue government-verifiable certificates (NCVET-aligned) with a unique ID and verification QR.	MUST	State, Citizen
FR-6.3	Certificates shall be pushed to DigiLocker and be independently verifiable online.	MUST	Citizen
FR-6.4	The system shall enforce certification SLA (e.g. assessment within 15 days of module completion) and flag breaches.	SHOULD	Nodal QC, District

4.7 Placement, Jobs & Recruiter Module

Req ID	Requirement	Priority	Role(s)
FR-7.1	The system shall provide profile-matched job discovery across government, vendor/empanelled, and private-sector openings.	MUST	Citizen
FR-7.2	The system shall aggregate private-sector openings from external boards (e.g. Naukri/LinkedIn/Monster) under their API terms, clearly attributed.	COULD	Citizen
FR-7.3	Recruiters shall access the certified talent pool, post jobs, and shortlist candidates with direct-apply routing.	MUST	Recruiter
FR-7.4	The system shall provide SDDI hiring insights: companies, districts covered, and hires by programme mix.	SHOULD	Recruiter, State
FR-7.5	The system shall surface cross-industry courses (Udemy/Coursera/upGrad/LinkedIn) and advancement pathways (ITI/diploma/degree/MBA).	COULD	Citizen
FR-7.6	The system shall publish sector salary benchmarks by experience level.	COULD	Citizen

4.8 Dashboards, Reports & Monitoring

Req ID	Requirement	Priority	Role(s)
FR-8.1	The system shall provide role-specific dashboards for all twelve user classes, enforced by RBAC.	**MUST**	All
FR-8.2	The system shall provide State and District dashboards with enrolment, certification, placement, and district-performance analytics.	**MUST**	State, District
FR-8.3	The MIS hub shall consolidate spoke (zone/district/centre) data into a single statewide source of truth with reconciliation.	**MUST**	MIS
FR-8.4	The system shall measure impact: trainer hours, attendance participation, officer-monitoring visits, and an impact score by centre/district/zone (N/S/E/W).	**MUST**	State, MIS, Data Science
FR-8.5	The system shall provide auto-generated, audit-ready reports (monthly progress, placement outcomes, district scorecards, financial utilisation).	**MUST**	State, District, MIS
FR-8.6	The Data Science team shall deploy predictive models (dropout-risk, placement-propensity, demand-forecast, anomaly detection) surfaced as advisory flags.	**SHOULD**	Data Science

4.9 Grievance Redressal

Req ID	Requirement	Priority	Role(s)
FR-9.1	Citizens shall raise categorised grievances that are auto-routed to the responsible cell with an SLA clock.	**MUST**	Citizen
FR-9.2	The system shall track grievances through Raised → Routed → In Progress → Resolved, with citizen-visible status.	**MUST**	Citizen, District
FR-9.3	Officers shall action grievances (take action, resolve, escalate) with full audit trail.	**MUST**	District, State, Nodal QC
FR-9.4	Tele-Customer Service agents shall log and update grievances on behalf of citizens calling the helpline.	**MUST**	Tele-CSR
FR-9.5	The system shall provide a grievance dashboard at citizen, district, state, and platform levels with category/priority/SLA analytics.	**MUST**	All officers

4.10 Software Helpdesk & Administration

Req ID	Requirement	Priority	Role(s)
FR-10.1	The system shall provide a software helpdesk/ticketing module scoped state-wise and district-wise.	MUST	Master Admin, State, District
FR-10.2	Tickets shall carry category, priority, level, and SLA, and progress through Open → Assigned → In Progress → Resolved.	MUST	Master Admin
FR-10.3	The Master Admin console shall provide user & role management, system health, and configuration.	MUST	Master Admin
FR-10.4	The system shall maintain an immutable audit & access log (user, action, module, IP, timestamp, geo) with defined retention.	MUST	Master Admin, Nodal QC
FR-10.5	Tele-CSR agents shall have a call-disposition and CRM interface linked to citizen records and tickets.	MUST	Tele-CSR
FR-10.6	The Nodal QC team shall run SOP-adherence audits and QC sampling across the hub-and-spoke network, raising non-conformances to closure.	MUST	Nodal QC

4.11 Notifications & Mobile App

Req ID	Requirement	Priority	Role(s)
FR-11.1	The system shall send push, SMS, and in-app notifications for sessions, certificates, jobs, grievances, and flags.	MUST	Citizen, Trainer
FR-11.2	The mobile app shall be offline-first for content and attendance, syncing on reconnection.	MUST	Citizen, Trainer, Onsite Supervisor
FR-11.3	The system shall support Gujarati, Hindi, and English across web and mobile.	MUST	All

4.12 DBT, Stipend & Financial

Req ID	Requirement	Priority	Role(s)
FR-12.1	The system shall process Aadhaar-seeded DBT stipend/scholarship disbursements via PFMS with status reconciliation.	MUST	State, District
FR-12.2	The system shall flag DBT/payment failures and route them to grievance/helpdesk.	MUST	District, Tele-CSR
FR-12.3	The system shall provide financial-utilisation reporting for audit.	MUST	State, MIS

5. External Interface Requirements

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5.1 User Interfaces

Responsive web application (desktop/tablet) and native Android/iOS apps. All interfaces shall be multilingual (Gujarati/Hindi/English), WCAG 2.1 AA compliant, and optimised for low-bandwidth rural use. UI shall follow Government of India Guidelines for Indian Government Websites (GIGW).

5.2 Hardware Interfaces

The system shall interface with camera (photo/document capture), GPS (geo-tagging of attendance and field reports), and, where deployed, biometric devices for attendance. Biometric hardware supply is out of scope; software shall support standard STQC-certified device SDKs.

5.3 Software Interfaces

Interface	Purpose	Protocol
UIDAI Aadhaar	e-KYC/authentication	REST via API Setu / AUA-KUA
DigiLocker	Document fetch, certificate push	REST/OAuth2
DBT / PFMS	Stipend disbursement, status	Secure API / SFTP as mandated
APAAR / ABC	Academic credit linkage	REST
Conferencing (Meet/Zoom/Jitsi)	Live class create/join/record	SDK / REST
SMS Gateway	OTP & notifications	DLT-registered gateway
Job boards (optional)	Private-sector openings	Provider APIs under commercial terms

5.4 Communication Interfaces

All external and client communication shall use TLS 1.2+ (HTTPS). Internal service communication shall be encrypted. APIs shall use OAuth2/JWT with short-lived tokens and refresh rotation.

6. Non-Functional Requirements

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6.1 Performance & Scalability

Req ID	Requirement	Priority
NFR-1.1	The system shall support at least 8.5 lakh (0.85M) registered citizens at launch, scaling to several million, across 33 districts.	MUST
NFR-1.2	Standard page/API responses shall complete within 2 seconds at the 95th percentile under expected load.	MUST
NFR-1.3	The system shall support at least 50, 000 concurrent active users with horizontal scalability.	MUST
NFR-1.4	Reporting/analytics workloads shall run against a separate store so as not to degrade transactional performance.	SHOULD

6.2 Security & Data Protection

Req ID	Requirement	Priority
NFR-2.1	The system shall comply with the DPDP Act 2023, Aadhaar Act, and CERT-In guidelines.	MUST
NFR-2.2	All data shall be encrypted in transit (TLS 1.2+) and at rest (AES-256).	MUST
NFR-2.3	The system shall pass CERT-In-empanelled VAPT prior to go-live and periodically thereafter.	MUST
NFR-2.4	Access shall be governed by least-privilege RBAC with mandatory MFA for privileged roles.	MUST
NFR-2.5	Raw Aadhaar numbers shall not be stored; only permitted reference tokens with consent artefacts.	MUST

6.3 Availability & Reliability

Req ID	Requirement	Priority
NFR-3.1	The system shall provide 99.5% uptime (target 99.9%) measured monthly, excluding scheduled maintenance.	MUST
NFR-3.2	The system shall implement automated backups with a defined RPO $\leq$ 15 min and RTO $\leq$ 4 hours.	MUST
NFR-3.3	The system shall degrade gracefully and queue offline field transactions for later sync.	SHOULD

6.4 Usability & Accessibility

Req ID	Requirement	Priority
NFR-4.1	The system shall meet WCAG 2.1 AA and GIGW standards.	MUST
NFR-4.2	Core citizen journeys shall be completable on a low-end Android device over 3G.	MUST
NFR-4.3	The UI shall be fully available in Gujarati, Hindi, and English.	MUST

6.5 Maintainability & Portability

Req ID	Requirement	Priority
NFR-5.1	The system shall be containerised and deployable via infrastructure-as-code for repeatable, auditable releases.	MUST
NFR-5.2	Code shall follow documented standards with automated testing and CI/CD gates.	MUST
NFR-5.3	The system shall avoid proprietary lock-in that would prevent migration between empanelled Indian clouds.	SHOULD

7. Data Requirements & Localization

Contents ↑

All personal and government data, including citizen profiles, documents, certificates, attendance, assessments, and financial records, shall be stored exclusively within India on MeitY-empanelled infrastructure. No personal data shall be transferred, processed, or backed up outside Indian jurisdiction.

- **Data classification:** data shall be classified (public / internal / confidential / sensitive-personal) with handling rules per class.
- **Consent & purpose limitation:** every personal-data use shall be tied to a captured, revocable consent artefact under the DPDP Act.
- **Retention:** retention periods shall be defined per data class; audit logs retained per CERT-In (minimum 180 days rolling, longer where mandated).
- **Anonymisation:** analytics and Data Science access shall use anonymised/aggregated datasets; re-identification shall be prevented by design.
- **Data subject rights:** the system shall support access, correction, and erasure requests consistent with the DPDP Act.

8. Security, Compliance & Audit

Contents ↑

Security is a first-class, non-negotiable requirement given the handling of Aadhaar-linked and minors' data.

- **Compliance:** DPDP Act 2023; Aadhaar Act & UIDAI regulations; CERT-In directions; NCVET/NSQF for certification; GIGW for accessibility.
- **Identity & access:** RBAC with least privilege across all twelve user classes; MFA for privileged roles; SSO where applicable.
- **Audit trail:** immutable logging of every privileged and data-changing action (user, action, module, IP, timestamp, geo).
- **Child-safety:** special handling for minors, restricted data exposure, guardian-consent workflows, and no content or contact features that could facilitate harm.
- **Security testing:** CERT-In-empanelled VAPT before go-live and at defined intervals; secure SDLC with code review and dependency scanning.
- **Incident response:** defined runbook and CERT-In incident reporting within mandated timelines.

9. Reporting & Analytics

Contents ↑

The MIS hub and Data Science team jointly own the analytics layer. Reporting shall include:

- **Operational MIS:** enrolment, attendance, certification, placement, by centre, district, zone (N/S/E/W), and sector, consolidated hub-and-spoke.
- **Impact measurement:** trainer hours, attendance participation, officer-monitoring coverage, and a composite impact score tying training → certification → placement → income.
- **Governance reports:** SOP-adherence (Nodal QC), grievance SLA performance, helpdesk SLA, DBT utilisation, audit-ready.
- **Predictive analytics:** dropout-risk, placement-propensity, sector demand-forecast, and anomaly/fraud detection, surfaced as advisory flags for officers.
- **Ministry briefings:** one-click auto-generated PDF/XLSX briefs for leadership.

10. Deployment & Infrastructure

Contents ↑

- **Hosting:** MeitY-empanelled cloud (GI Cloud / MeghRaj or empanelled CSP), India region only.
- **Environments:** separate Development, Staging/UAT, and Production, with data isolation.
- **Release strategy:** containerised, IaC-driven, blue-green/rolling deploys with rollback.
- **Observability:** centralised logging, metrics, alerting, and uptime monitoring feeding the Master Admin console.
- **Business continuity:** automated backups, cross-AZ redundancy, and a documented DR plan meeting the stated RPO/RTO.

11. Assumptions, Constraints & Risks

Contents ↑

Risk	Impact	Mitigation
Delayed sandbox/production access to India Stack APIs (Aadhaar, DBT, DigiLocker).	High	Early engagement with authorities; mock adapters to parallelise development; schedule allowance on integration paths.
Rural connectivity gaps impacting field usage.	Medium	Offline-first mobile layer; SMS fallback; low-bandwidth optimisation.
Scope creep from evolving departmental requirements.	High	MoSCoW prioritisation; formal change-control with re-baselining.
Data-protection / security non-conformance.	High	Compliance-first design; CERT-In VAPT; DPDP governance from day one.
Third-party feed (job boards/course platforms) API/ commercial changes.	Low	Treated as optional/attributed; graceful degradation if unavailable.
Change of trained field staff / adoption resistance.	Medium	Training, multilingual UX, tele-support helpline, onsite supervisor support.

12. Effort & Delivery Estimation

Contents ↑

12.1 Estimation Methodology

Effort is estimated bottom-up by module, expressed in **person-months (PM)**, and aggregated into a delivery plan. Estimates reflect the full engineering scope of a population-scale, compliance-heavy government platform, including India Stack integration, security auditing, multilingual and offline-first delivery, and the twelve operational roles specified in this document.

12.2 Estimated Effort by Module

Module / Workstream	Effort (PM)	Share
Identity, Onboarding, Profile & India Stack e-KYC	11.7	8.1%
Document Vault, DigiLocker & Consent (DPDP)	7.8	5.4%
Training Lifecycle & Enrolment	6.5	4.5%
Trainer & Batch Management	5.9	4.1%
Content Library & LMS (authoring/localisation workflow)	8.5	5.9%
Live Classes, Recording & Attendance (offline-first)	9.1	6.3%
Assessment & Certification (DigiLocker, NCVET)	7.2	5.0%
Placement, Jobs & Recruiter Module	7.8	5.4%
Dashboards, MIS Hub-and-Spoke & Reporting	10.4	7.2%
Impact Monitoring & Data Science (models/pipelines)	8.5	5.9%
Grievance Redressal	4.5	3.1%
Software Helpdesk & Master Admin Console	6.5	4.5%
Tele-CSR / CRM & Call-Disposition	4.5	3.1%
Nodal QC / SOP-Adherence & Audit	4.5	3.1%
Notifications & Mobile Apps (Android/iOS)	10.4	7.2%
DBT / PFMS Stipend & Financial	6.5	4.5%
Security, Compliance, VAPT remediation & DPDP	7.8	5.4%
Platform: infra, CI/CD, IaC, integration harness	7.8	5.4%
QA, UAT, documentation & training	7.8	5.4%
Total Estimated Effort	**143.7 PM**	**100%**

12.3 Indicative Team & Timeline

Delivery is resourced by a **blended team of approximately 150 people** spanning two wings: a core engineering and specialist build team (Section 16.1) and an operations wing (Tele-CSR/contact centre, MIS hub-and-spoke, onsite field supervisors, Nodal QC, and content production) that stands up alongside the platform and continues into steady-state operations. This larger, parallelised team is what compresses delivery: the estimated 143.7 PM of engineering is executed across concurrent module workstreams to reach **full production go-live within approximately 9 calendar months**, with a compliant core available substantially earlier under the phased plan in Section 13. The schedule carries deliberate allowance on the two dependencies that sit on the critical path and are not reducible by headcount, India Stack sandbox-to-production access and CERT-In security certification (VAPT), which are sequenced early and tracked as programme risks.

> **Basis of estimate.** Figures are indicative planning estimates that frame the engagement. Final, contractually-binding effort and cost are established during detailed design with the appointed development partner, using this SRS as the baseline.

13. Project Phasing & Milestones

Contents ↑

Phase	Key Deliverables	Window
Phase 0, Mobilisation	Team onboarding at scale, environment setup, India Stack sandbox access initiated, detailed design, security baseline.	M0, M1
Phase 1, Identity & Core (compliant core live)	Onboarding, e-KYC, profile, document vault, RBAC for all roles, admin console; first compliant production slice.	M1, M3
Phase 2, Delivery & Content	Training lifecycle, trainer/batch, content LMS, live classes, attendance (offline-first).	M2, M5
Phase 3, Certification & Placement	Assessment, certification (DigiLocker), placement, recruiter, DBT stipend.	M4, M6
Phase 4, MIS, Monitoring & Governance	Hub-and-spoke MIS, dashboards, impact analytics, grievance, helpdesk, Tele-CSR, Nodal QC.	M5, M7
Phase 5, Hardening & Full Go-Live	VAPT clearance, UAT, performance testing, training, data migration, full production go-live of all modules.	M7, M9
Phase 6, Stabilisation & O&M	Warranty, SLA operations, iterative enhancements under change-control.	Post go-live

Phases run concurrently across the blended team (Section 12.3); a compliant core is live by ~M3 and all modules by ~M9. Windows retain allowance on the integration and certification paths, which are sequenced early because they are not compressible by headcount.

14. Acceptance Criteria & Exit

Contents ↑

The system shall be accepted when the following are demonstrably met:

- All **MUST** functional requirements (Section 4) are implemented and pass UAT for every applicable user class.
- All non-functional targets (Section 6) are met, evidenced by performance and load test reports.
- A CERT-In-empanelled VAPT is passed with no open critical/high findings; DPDP and Aadhaar compliance are evidenced.
- India Stack integrations (Aadhaar e-KYC, DigiLocker, DBT/PFMS) are live and reconciled in production.
- Hub-and-spoke MIS reconciles to a single statewide source of truth; audit logs are complete and immutable.
- Multilingual (Gujarati/Hindi/English) and WCAG 2.1 AA / GIGW accessibility are verified.
- Training completed for all operational roles; documentation and runbooks delivered.

Each requirement in Section 4 maps to a test case; a requirements-traceability matrix (RTM) shall accompany UAT sign-off.

15. Appendices

Contents ↑

Appendix A, User Classes Summary

Twelve RBAC user classes: State Skill Mission/SOP; District Skill Officer; Trainer; Recruiter; Citizen/Trainee; Software Master Admin; Tele-Customer Service Agent; MIS Team (Hub & Spoke); Onsite Supervisor (Field); Nodal QC & SOP-Adherence Team; Content & Creative Team; Data Science Team.

Appendix B, Compliance Checklist

- [ ] DPDP Act 2023, consent, purpose limitation, data-subject rights
- [ ] Aadhaar Act / UIDAI e-KYC, no raw Aadhaar storage
- [ ] MeitY-empanelled Indian cloud, data localization
- [ ] CERT-In, VAPT, audit-log retention, incident reporting
- [ ] NCVET / NSQF, certification alignment
- [ ] WCAG 2.1 AA / GIGW, accessibility & multilingual
- [ ] Child-safety, minors' data handling & guardian consent

Appendix C, Traceability

A full Requirements Traceability Matrix (requirement → design → test case → acceptance) shall be maintained by the development partner and delivered at UAT.

16. Project Resourcing

Contents ↑

The delivery organisation is a **blended team of approximately 150 people** across two wings: a **core engineering and specialist build team** (Section 16.1) and an **operations wing** (Section 16.2) that stands up alongside the platform and continues into steady-state managed operations. Headcount reflects the full engineering, security, data, content, and quality scope described in this document, together with the operational staffing needed to run the contact centre, hub-and-spoke MIS, field verification, and quality functions. Staffing is phased (Section 13): the engineering wing peaks during build and hardening, while the operations wing scales up toward go-live and sustains through the support term. This delivery organisation is distinct from the twelve platform *user* roles defined in Section 2.3.

16.1 Core Delivery Team, Headcount, Qualification & Experience

Role	Count	Qualification	Experience
Project / Delivery Manager	1	PMP/Prince2; B.E./MBA	12+ yrs, govt/large IT programmes
Solution / Enterprise Architect	1	B.E./M.Tech, Computer Science	12+ yrs, DPI/large-scale systems
Backend Engineers	6	B.E./B.Tech/MCA	3, 8 yrs, Python/Node, PostgreSQL
Frontend Engineers (Web)	4	B.E./B.Tech/MCA	3, 7 yrs, React/Next.js
Mobile App Developers	3	B.E./B.Tech/MCA	3, 7 yrs, React Native/Flutter, offline-first
Hybrid Cloud & Network Specialists	2	B.E./B.Tech; cloud/network certs (CCNP/CKA)	6, 10 yrs, hybrid cloud & networking
Hybrid Cloud Storage Engineers	1	B.E./B.Tech; storage/cloud certs	6, 10 yrs, object/block storage, backup/DR
Middleware / Integration Engineers	2	B.E./B.Tech/MCA	5, 9 yrs, API/ESB, India Stack integration
Cybersecurity Engineers	2	B.E./B.Tech; CISSP/CEH/CISA	6, 10 yrs, appsec, VAPT, DPDP/ CERT-In
DevOps / SRE Engineers	2	B.E./B.Tech	5, 9 yrs, CI/CD, IaC, containers
Database / Data Engineers	2	B.E./B.Tech/MCA	5, 9 yrs, PostgreSQL, pipelines
Data Scientists	2	M.Sc./M.Tech, Statistics/CS/ML	4, 8 yrs, ML, analytics
QA / QC Engineers	4	B.E./B.Tech/MCA; ISTQB	3, 8 yrs, manual + automation, UAT
UI/UX Designers	2	B.Des/Graphic/HCI	4, 8 yrs, accessibility, multilingual UX
Content & Creative Producers	3	Graduate; media/animation	3, 7 yrs, video/infographic/animation
Business Analyst / MIS Lead	2	B.E./MBA	5, 9 yrs, requirements, MIS/reporting
Compliance / Data-Protection Officer	1	LL.B./compliance cert	8+ yrs, DPDP/privacy/audit
Support / Tele-CSR Leads	2	Graduate	4, 7 yrs, service desk/contact centre
Engineering & Specialist Build Wing	**42**	**Peak concurrent headcount, build and hardening phases**	

16.2 Operations Wing, Headcount & Function

Function	Count	Role & scope
Tele-CSR / Contact-Centre Agents	48	Inbound/outbound citizen helpline, multilingual (Gujarati/Hindi/English), grievance and ticket logging, call disposition.
MIS Hub-and-Spoke Operators	22	Central hub plus zonal spokes (N/S/E/W); data consolidation, reconciliation, SOP and target dissemination, reporting.
Onsite Field Supervisors	18	Centre-level verification, geo-tagged field reports, learner presence and infrastructure checks.
Nodal QC & SOP-Adherence Officers	10	QC sampling of batches/trainers/centres, SOP-conformance audits, non-conformance tracking to closure.
Content Production & Localisation	6	Multilingual course content, creative visuals, infographics, animations, publishing workflow.
Operations Support & Coordination	4	Roster, shift, and SLA coordination across the operations wing.
Operations Wing	**108**	**Scales toward go-live; sustains through the support term**

> **Total blended delivery organisation: approximately 150 people** (42 engineering & specialist + 108 operations), phased per Section 13.

> **Resourcing note.** Headcount is indicative and phase-dependent; not all roles are engaged for the full duration. Specialist roles, hybrid cloud & network, hybrid cloud storage, middleware/integration, cybersecurity, and QC, are weighted to the integration, hardening, and go-live phases. Operations & maintenance staffing (helpdesk, Tele-CSR, SRE, security) continues post go-live under the support engagement.

16.3 Specialist Capabilities

- **Hybrid Cloud & Network Specialists**, design and operate the MeitY-empanelled hybrid cloud footprint, VPC/network segmentation, connectivity to government networks, and load/traffic management.
- **Hybrid Cloud Storage Engineers**, object/block storage, encryption-at-rest, backup, archival, and disaster-recovery meeting the stated RPO/RTO with data localisation.
- **Middleware / Integration Engineers**, API/ESB layer, India Stack connectors (Aadhaar, DigiLocker, DBT, APAAR/ABC) via API Setu, and inter-module messaging.
- **Cybersecurity Engineers**, secure SDLC, application security, VAPT remediation, DPDP/CERT-In compliance, incident response, and audit-log integrity.
- **Mobile App Developers**, Android/iOS apps with offline-first field capability, push notifications, and low-bandwidth optimisation.
- **QA / QC Engineers**, functional, regression, performance, and accessibility testing; automated test suites; UAT coordination and requirements traceability.

> **End of Software Requirements Specification, SDDI Platform, v1.0**
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